Experiment No: 08

Experiment Name: Experiment on Frequency Modulation & Frequency Demodulation using Hardware.

Objective:

- To understand how to generate FM modulation.
- To observe the changes in FM signals in different parameter.
- To observe the signal reconstruction by FM demodulation.

Requirement:

1. Module KL-93004
2. Oscilloscope – 1p
3. Function generator – 2p
4. Power Supply
5. Connecting Wire & Plug

Circuit Diagram:

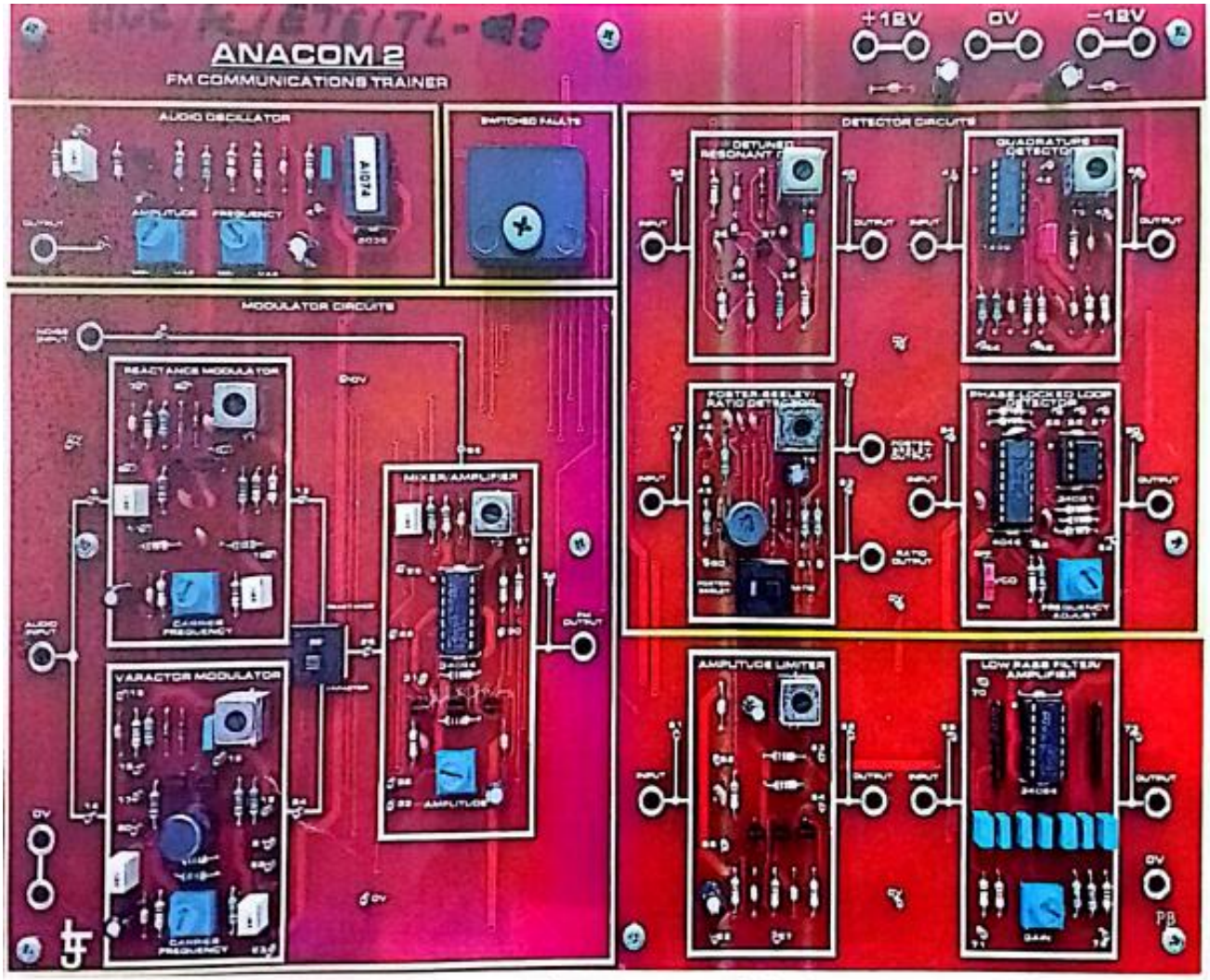


Figure 1. FM Modulation & Demodulation Circuit.

Working Procedure:

Follow the following steps:

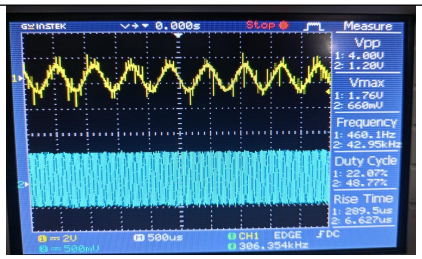
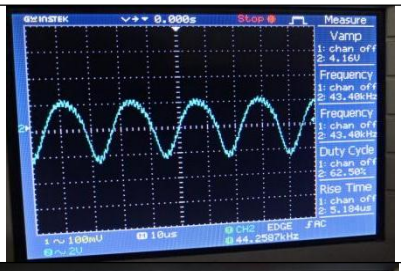
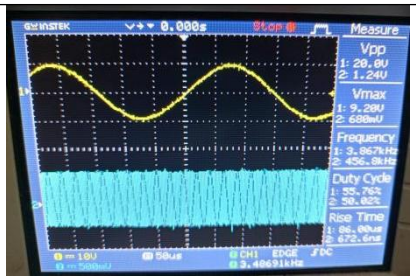
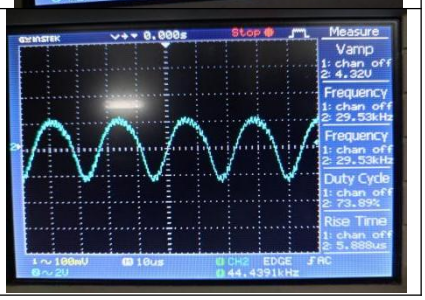
1. Locate LM 566 IC in FM modulator KL-93004.
2. Insert coupling plug J3 to set capacitor 0.01 uf.
3. Observe the output signal using oscilloscope. Record the value of Fc in table 1.
4. Set DC power supply voltage 3.6v and connect it at pin 5. By adjusting VR1 set the output at 20KHz.
5. Record Table 1 according to given value. You should be careful during controlling DC output voltage. Never exceed the voltage above 7 volts.

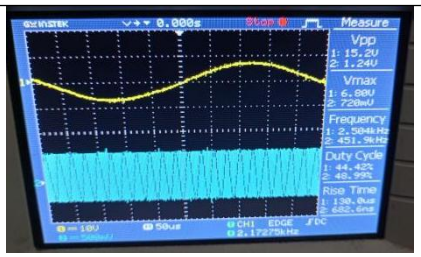
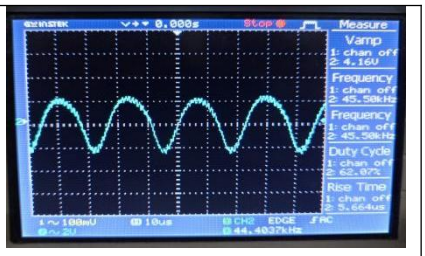
Data Table 1

Input Voltage	4v	15v	20v
Output Frequency	306.354KHz	2.172KHz	3.406KHz

6. Now remove DC power supply, insert J1.
7. Locate PLL circuit for demodulation. Insert J1 and J3.
8. We will observe Modulation and Demodulation simultaneously as like as previous lab classes.
9. Set the output of Function generator according to given value in Table 2 and connect Function Generator at the input of modulator. Record the data in Table 2 for different input signal.

Data Table 2

Input Parameter				**Modulated Output Waveform**	**Demodulated Output Waveform**
Vc	**Vm**	**Fc**	**Fm**		
	500 mv	**200 KHz**	**1KHz**		
	1v	**20KHz**	**1KHz**		

	2v	**300 KHz**	**2KHz**		

Discussion: In this experiment, we explored frequency modulation and demodulation using the KL-93004 hardware module. By varying the DC input voltage to the LM566 IC, we observed clear changes in carrier frequency, showing how FM generation depends on control voltage. The recorded data confirmed that higher voltages produced lower carrier frequencies, emphasizing the sensitivity of FM signals to input parameters. Using the PLL circuit, the demodulated output successfully reconstructed the original message signal, such as the 1 kHz input, proving the accuracy of FM demodulation. Overall, the experiment demonstrated the practical relationship between modulation parameters and signal recovery, reinforcing theoretical concepts of FM communication systems.